

SOLAPUR, India

WMO: 431170

Lat: 17.628N

Lon: 75.935E

Elev: 1584

StdP: 13.87

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.5	62.1	36.8	34.0	79.9	40.0	38.5	79.0	5.5	78.2	5.3	78.2	0.8	50	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.6	106.3	72.1	104.5	72.1	102.7	72.0	79.6	91.4	78.4	89.6	77.6	88.6	2.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	149.2	84.3	75.9	143.8	83.2	75.0	139.3	82.3	44.7	91.7	43.4	89.5	42.5	88.3	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.8	5.5	5.2	DB	54.1	109.3	2.3	1.4	52.4	110.3	51.1	111.1	49.8	111.9	48.1	112.9
			WB	47.6	83.1	1.7	1.9	46.4	84.5	45.4	85.6	44.4	86.6	43.2	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	76.0	80.9	86.9	91.7	91.9	85.3	81.7	81.0	81.1	81.0	78.3	75.6	(6)
	DBStd	5.98	2.96	3.15	3.70	3.21	3.30	3.83	2.71	2.30	2.37	2.57	2.65	3.02	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11908	806	866	1145	1251	1300	1058	983	961	934	960	850	794	(10)
	CDD65	6433	341	446	680	801	835	608	518	496	484	495	400	329	(11)
	CDH74	76829	3126	5023	9582	12503	12861	7473	5208	4688	4577	4957	3854	2977	(12)
(13)	CDH80	40471	1273	2590	5796	8320	8497	3751	1944	1661	1720	2150	1621	1148	(13)
(14)	Wind	WSAvg	2.0	1.6	1.7	1.8	2.0	2.5	2.8	2.7	2.3	1.9	1.5	1.4	(14)
(15) Precipitation	PrecAvg	28.2	0.1	0.3	0.2	0.4	1.2	4.2	5.0	4.5	7.8	3.5	1.0	0.2	(15)
	PrecMax	43.5	1.1	2.0	1.2	1.9	5.1	10.1	10.9	9.6	18.2	10.9	6.6	1.4	(16)
	PrecMin	12.0	0.0	0.0	0.0	0.0	0.0	0.7	0.9	0.0	0.8	0.0	0.0	0.0	(17)
	PrecStd	8.2	0.2	0.6	0.3	0.6	1.3	2.1	2.7	3.0	4.8	3.2	1.5	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	98.7	104.9	108.2	108.9	104.3	95.0	93.1	94.3	95.5	93.0	92.1	(19)
		MCWB	67.8	68.1	70.6	72.2	73.1	74.6	75.2	74.9	74.2	71.8	69.1	69.0	(20)
	2%	DB	90.5	96.5	102.7	106.3	106.9	100.3	92.5	91.2	92.0	93.2	90.9	89.4	(21)
		MCWB	67.1	67.8	70.6	71.8	72.7	75.3	75.0	74.6	73.8	70.8	68.8	67.7	(22)
	5%	DB	88.7	94.5	100.9	104.7	104.9	97.2	90.7	89.6	90.2	91.4	89.5	87.6	(23)
		MCWB	66.6	68.0	70.1	71.8	72.4	75.4	74.9	74.4	73.8	70.7	68.9	66.6	(24)
	10%	DB	86.8	92.2	98.8	102.9	102.8	94.2	88.7	87.8	88.4	89.7	88.0	86.1	(25)
		MCWB	66.2	67.8	69.9	71.9	72.5	75.3	74.8	74.4	74.0	70.8	68.5	66.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	78.5	81.0	81.0	81.1	80.5	79.3	79.3	78.6	78.4	76.6	73.6	(27)
		MCDB	86.1	90.3	99.3	100.5	94.9	90.9	86.9	86.3	86.3	84.5	86.7	83.7	(28)
	2%	WB	70.9	74.8	76.4	78.1	78.8	79.0	77.9	77.8	77.4	77.2	74.6	71.9	(29)
		MCDB	83.8	88.9	95.2	96.2	92.4	90.0	85.8	85.2	85.5	84.0	84.6	83.2	(30)
	5%	WB	69.3	71.9	73.7	76.5	77.5	77.9	76.9	76.8	76.5	76.2	73.3	70.5	(31)
		MCDB	83.5	88.2	93.9	95.5	92.0	89.2	84.6	84.3	84.7	83.5	83.1	82.5	(32)
	10%	WB	67.9	69.9	71.9	75.0	76.4	76.9	76.0	75.8	75.7	75.2	72.1	69.1	(33)
		MCDB	83.1	87.1	92.1	94.4	91.8	88.5	84.2	83.3	83.8	82.9	81.8	81.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	23.0	23.4	22.6	21.9	16.2	13.0	13.1	14.1	17.6	20.6	22.7	(35)
		MCDBR	24.4	25.0	25.2	23.9	23.7	20.7	16.4	16.2	16.9	21.0	22.6	24.2	(36)
		MCWBR	10.6	10.5	10.1	7.9	6.6	5.4	4.4	4.5	5.1	6.7	8.5	10.3	(37)
	5% WB	MCDBR	20.6	21.7	22.8	21.4	21.1	16.4	13.1	13.1	13.9	14.1	15.9	18.7	(38)
		MCWBR	10.4	12.0	11.5	9.4	7.2	5.8	5.1	5.6	5.5	5.9	7.2	9.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.447	0.462	0.535	0.596	0.588	0.558	0.534	0.518	0.507	0.535	0.487	0.452	(40)
	taud		2.095	2.079	1.902	1.815	1.851	1.980	2.058	2.105	2.131	1.992	2.061	2.103	(41)
	Ebn at Noon		263	266	251	236	235	240	246	252	255	242	249	257	(42)
	Edh at Noon		49	52	64	70	67	58	54	51	50	56	50	47	(43)
(44) All-Sky Solar Radiation	RadAvg		1593	1856	2050	2156	2164	1648	1355	1375	1517	1617	1553	1519	(44)
(45)	RadStd		56	62	95	75	78	137	118	76	106	79	93	55	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	+0.97	+0.85	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page